Contents

Overview of the project

According to the World Health Organization, breast cancer is one of the leading causes of death among women. It was the most common cancer in women in 157 countries out of 185 in 2022. 670,000 women died from this disease worldwide in the same year. Statistically, over the course of their lives, 1 woman out of 12 will suffer from breast cancer, making a compelling case for prevention. Although the techniques may vary, one of the most prevalent is mammography. It is a well-established imaging technique that has undergone significant improvements ranging from digital mammography to digital breast tomosynthesis [1]. Historically, mammograms were interpreted exclusively by specialists, but CNN-based methods are increasingly used to support diagnostic workflows.

Relating to the template 3.2. "Deep Learning Breast Cancer Detection", the project aims to classify suspicious mammographic abnormalities as benign or malignant. Mammography assessment can be divided into two main tasks: identifying abnormalities in the image, known as computer-aided detection (CADe), and assessing whether an identified abnormality is benign or malignant, known as computer-aided diagnosis (CADx). While CADe may prompt further investigation, CADx supports the diagnostic characterisation of an identified abnormality. This project focuses on the latter.

Traditionally, this assessment is performed by radiologists, usually two of them for one patient. Their assessments provide an early indication of the nature of the tumour, but they are subject to inter-reader variability, fatigue, workload and differences in expertise. Computer-aided diagnosis has developed partly to address these inherent limitations.

Originally, CAD relied on traditional machine-learning approaches. Tahmasbi et al. [2] achieved high accuracy by using Zernike moments. Their approach does not require a large dataset - 322 digitized mammograms, in this case. In a field where data is scarce and of variable quality, this is an upside. However, traditional machine learning models require manual feature selection, which is also human-dependent and prone to biases. For instance, two selected features can be highly correlated with each other, causing unnecessary noise.

The promise of deep learning lies in its ability to learn relevant patterns with less manual feature engineering. Such learning can also be accelerated thanks to advances in computing hardware. Wang et al. mentions the crucial role of GPU and TPU [1]. This project consists of developing a CNN-based model. When it comes to image analysis, CNNs are particularly well suited to computer-vision tasks. However, in the case of CADx on mammograms, they also come with challenges.

Wang et al. insists on the importance of using large datasets for DL models to properly learn. Unfortunately, mammograms are rare and curated ones are rarer, raising some questions.

- Should an extensive dataset be built by compiling several sources of data?
- Can the limitations of a relatively small dataset be mitigated through model design and regularisation?

What is more, the study pinpoints the risks incurred by human annotations. A deep-learning model may learn from such misleading artefacts, making a case for an enhanced preprocessing.

Further to these concerns, it is worth adding that deep learning techniques can be opaque despite their promises. Their predictions therefore require careful interpretation. In the medical field, these algorithms deal with human lives and explaining a machine decision is a critical matter. The model's performance should also be evaluated according to sensible metrics. In this context, recall and precision need to be separately evaluated. Sensitivity is particularly important because the consequences of missing a malignant lesion may be more serious than those of incorrectly flagging a benign lesion.

Literature review

CBIS-DDSM Dataset

CADx requires a reliable dataset for training. Such data are scarce. Varoquaux and Cheplygina emphasise that available datasets suffer from several biases. Although it is a common practise in machine learning to build a test set that is a subset of the same data pool as the train set, there is a risk of validating a model's performance on a population that does not reflect the distribution of the actual population, causing poor performance of the model in real-life situations. In addition, a dataset may not necessarily capture the full extent of medical conditions, leading to model underperformance if it was trained on data underestimating them. The dataset also requires high-quality and curated images. Indeed, they can sometimes be affected by traces of medical intervention, leading the model to learn inappropriate features. Besides, the labelling needs to be accurate and of high quality. Finally, the set should not contain any patient leakage because it can overestimate the obtained result by allowing the model to learn patterns that will be found in the test set [3].

Being aware of these shortcomings, this study uses the CBIS-DDSM reference dataset. It is a curated version of the DDSM dataset, which is a collection of mammograms from different sources, "Massachusetts General Hospital, Wake Forest University School of Medicine, Sacred Heart Hospital, and Washington University of St Louis School of Medicine" [4]. To some extent, the CBIS-DDSM dataset represents an attempt to tackle the biases the legacy DDSM faced as described by Lee et al. First, the accessibility of data is improved. The metadata are provided in a single CSV file, including patient characteristics (BI-RADS, age, etc). Some mass cases were also questionable as to whether they represented genuine signs of disease. With the help of a trained mammographer, 339 images were removed that could not be appropriately confirmed. Considering the original DDSM dataset accounted for 2620 scanned film mammography studies, this removal represents a substantial proportion of the original dataset and reduces the risk of training on uncertain cases. The CBIS-DDSM dataset also comes with an enhanced region-of-interest (ROI) segmentation for mass cases. The original DDSM provided a mere location of the mass tumour, which is not sufficient for a model to properly train. By using the original ROI as a rough starting point, the region was further cropped to the exact contour using Chan-Vese model [5]. Finally, the CBIS-DDSM also comes up with a standardized train/test split. For mass cases, on which this study focuses, the train set accounts for 51.37% benign cases and 48.63% malignant cases. The test set, on the other hand, is 58.5% benign and 41.5% malignant.

Critical assessment of literature

Architectures

As previously stated, the CBIS-DDSM is a bank of medical cases. A case indicates a certain abnormality. The medical process entails two mammograms per abnormality, the craniocaudal (CC) and the mediolateral oblique (MLO) views. The CBIS-DDSM dataset also comes with a cropped image of the region and a mask for the full image. Every patient examination may thus come up with four documents. This structure therefore enables several preprocessing strategies. Liao and Aagaard converts all DICOMs into .png. The authors built a pipeline that consists in applying the mask to the full mammogram and that centers around the tumour. The mask is strict, meaning that pixels outside the region of interest are set to black. Ruling out the default cropped images allows the authors to control the preprocessing pipeline rather than relying directly on the provided cropped images. As a prerequisite, the mask dimensions should match those of the mammogram. Otherwise, the case is excluded because it is not considered reliable [6]. The main limitation of a strict ROI-masking approach is that it may remove contextual information surrounding the lesion.

In contrast to the lesion-focused approach of Liao et al. highlight the importance of contextual information in mammographic diagnosis., Abdikenov et al. emphasises that tumour diagnosis occurs in a contextual analysis of the full mammogram. The presence of calcifications - compatible with a malignant form - the breast density, whether it is a CC or a MLO view orient the radiologist's judgment. This information can only be appreciated by assessing the full mammogram, not just the abnormality. Their study concludes that training two models, one on CC views and the other one on MLO views, and, then, merging their features, gives better results than training a model on a merged image of both CC and MLO views [7]. As such, it emphasises the importance of the context. The architecture can somehow perform well under some models, like VGG19 (accuracy : 0.8865 and F1 : 0.6783) or DenseNet (accuracy : 0.7152 and F1 : 0.7582), but direct image merging may blur the signal and prevent generalisation. According to the authors' findings, two distinct models generalize better on heterogeneous data when trained on well-defined views.

The approach does not consider any crop though, meaning the tumour assessment is achieved in the context of the entire mammogram. The mammogram is considered as a full input and, as such, there is no notion of region of interest. A full-image or multi-view approach may capture contextual information, but it may also make it harder to isolate the contribution of the lesion itself because the patterns are learnt from an image containing a lot more information than the presence of a mass. There is a risk that the model learns global correlation associated with malignancy rather than lesion-specific features. Furthermore, depending on whether the mammograms were preprocessed, they may display human artifacts or traces of handwritting, which could introduce shortcut learning or spurious correlations. That is why Shen, et al. [8] adopts a hybrid approach combining a full mammogram view and ROI study called GMIC. Although the model trained at the full mammogram level has a CADe purpose (detection of an abnormality), its features are combined with those produced by the local module, encouraging the CADx process to acknowledge the full context. This study generates performance comparable to radiologists under the conditions evaluated while requiring less GPU memory and being 4.3 times faster than ResNet-34.

Source	**Main contribution**	**Relevance to this project**	**Limitation / gap**
Liao et al.	Propose a transparent CBIS-DDSM pipeline using ROI masks applied to full mammograms. Pixels outside the ROI are set to black, creating a strict lesion-focused representation.	Supports the use of hard ROI masking as a controlled preprocessing strategy for CADx. It helps the CNN focus on the annotated abnormality rather than unrelated image regions.	Strict masking may remove useful surrounding context, such as lesion margins, nearby tissue patterns, or breast density information.
Abdikenov et al.	Compares multi-view mammography strategies using CC and MLO views. Their results suggest that processing views separately before feature fusion can outperform simple image merging.	Supports the idea that mammographic context and view-specific information matter for classification. It motivates looking beyond isolated lesion crops.	Their approach uses full-view information and does not directly test ROI masking or lesion-centred representations. It therefore does not isolate the effect of preprocessing around the abnormality.

Source	Main contribution	Relevance to this project	Limitation / gap
Shen et al. / GMIC	Proposes a global-local architecture that analyses the full mammogram, selects informative local patches, and fuses global and local features for prediction.	Strongly supports the idea that both local lesion details and global mammographic context may contribute to classification. This motivates a hybrid or comparative approach.	GMIC is a complex weakly supervised architecture designed for high-resolution screening mammography. It is not a direct controlled comparison of crop, mask, and contextual ROI representations on CBIS-DDSM mass CADx.

Abdikenov et al. reduce the effective dataset size by pairing CC and MLO views. The CBIS-DDSM dataset primary key is the case, an aggregation of the patient id, the view and the nth occurrence the exam was conducted under these conditions. The study entails a reduction of the training set by consolidating CC and MLO views per patient, per occurrence. Since paired views are not available for every case, it may account for a significant loss of information. CNN training generally requires sufficiently large and representative datasets. Although this study acknowledges the benefit of a multi-angle approach, it is therefore less suitable for the present dataset. Shen et al. offer a solution by introducing a full context / cropped dichotomy. This applies at case level and provides complementary global and local information to the classifier. However, it entails an enhanced preprocessing due to the cropping engineering. Zooming over the lesion raises questions about the margins and the image resolution.

This review identifies two connected research questions. First, how does lesion representation - full mammogram or ROI-centred crop - affect benign–malignant classification under a controlled CNN baseline? Second, once an effective local representation has been selected, does combining it with a lightweight global branch improve performance over either input alone?

Evaluation

Varoquaux and Cheplygina stress that medical imaging studies require meaningful baselines, because performance claims are difficult to interpret if the model is not compared against an appropriate reference [3]. This is particularly relevant here, since the project does not only aim to train a CNN, but to compare different representations of the same lesion. A baseline is therefore needed to determine whether masking, contextual cropping, or global-local inputs actually improve performance. Chollet's deep learning workflow provides a practical framework for implementing a binary classification model. It consists of beginning with a simple working model, monitoring validation performance, diagnosing overfitting or underfitting, and only then increasing model complexity.

Baseline selection is not the only methodological issue ; metric choice also affects how model performance is interpreted. Accuracy is frequently used as a primary metric to assess the model's performance. Abdikenov et al. emphasise this metric when presenting a model. For instance, their study emphasizes that "the accuracy rose to 88.53%, with a remarkably high ROC AUC of 0.9283. These results suggest that the DBE architecture may better exploit inter-view feature complementarities or capture latent patterns in datasets." Although it is an interesting metric, false negatives and false positives do not carry the same clinical consequences in the medical profession. As a result, Varoquaux and Cheplygina insist that "important metrics may be missing from evaluation. Next to typical classification metrics (sensitivity, specificity, area under the curve), several authors argue for a calibration metric that compares the predicted and observed probabilities".

For this reason, this project should not rely on accuracy alone. AUC is useful for threshold-independent comparison, while recall/sensitivity is particularly important because false negatives are more problematic in a diagnostic setting. Precision remains relevant because excessive false positives may reduce clinical usefulness.

Overall, the reviewed literature suggests that CNN performance in mammographic CADx depends not only on architecture, but also on dataset quality, leakage control, lesion representation, contextual information, baseline choice, and metric selection.

Design

Domain and users

In the project scope, the presence of an abnormality is already known and needs to be assessed. The primary users would be radiologists who would use it as a decision-making tool.

From this perspective, the user and the target population should be differentiated. Assessing the nature of a tumour is a radiologist's job. Such a model would support specialists by providing a second-opinion signal, helping prioritise uncertain cases, and flagging predictions that require closer review. Their judgment remains decisive. However, the pipeline could assist with triage while supporting the review of uncertain cases. The level of human oversight should remain high, particularly for uncertain or borderline predictions. As a result, such pipeline needs to provide transparency and explainability to point towards cases that require further examination. Rezazade Mehrizi et al. emphasize the importance of decision system evaluation. AI suggestions can affect radiologists' decisions in mammography examination, making explainability and careful human-AI interaction central to clinical deployment. Indeed, when they do not spend extensive time investigating a mammogram, radiologists may be influenced by an incorrect AI recommendation. Some explainability tools mitigate this bias, especially visual heat maps [9].

Although breast cancer can also occur in men, male cases account for only around 0.5–1% of breast cancer cases. The target population represented by mammography screening datasets is therefore predominantly female.

Dataset

The model is trained and tested on the CBIS-DDSM dataset. Each abnormality is associated with several DICOM files ("digital imaging and communications in medicine") : a cropped image centred on the lesion, a full mammogram, and a binary mask locating the ROI.

These three files may be available twice for each case, one for the CC view, one for the MLO view. Therefore, there is a risk of a patient having one view in the train set and the other one on the validation set. Varoquaux and Cheplygina stress the importance of preventing patient leakage to avoid inflating the performance of the model [3]. No set should overlap based on the patient reference. To prevent the risk of leakage, train, validation and test sets will be confirmed as independent at patient level. The split is then fixed and applied to all runs for reproducibility purposes.

Data pre-processing

Preprocessing is not a neutral stage of the pipeline. Should the tumour be used as input alone or some context be associated to it? Should this context be blurred to orient the model toward the region of interest ("ROI") or should it remain unmodified?

Most studies do not consider the default cropped image as a candidate for training. Indeed, there is no control over the margin around the tumour and, as a result, the crop margin and field of view cannot be controlled consistently. Shen et al. apply a "greedy crop" that identifies candidate tumour regions from activation maps associated with image-level labels. However, we choose to use the provided binary mask to derive the ROI coordinates. This project is strictly limited to CADx task. Cropping using masking coordinates is a non-greedy algorithm that is more spatially precise than weakly supervised localisation because it uses an expert-provided ROI mask.

Cropping entails a trade-off between resolution and generalization. As Liao et al. put it, “while downsizing could alleviate some computing resource limitations, as performed in other studies, it may also result in the loss of important information”. The authors apply a generic margin by calculating the geometric center of the lesion and standardizing the margins from there. This study standardises crop dimensions to improve the consistency and generalisability of the input representation.

The geometric center of the lesion is calculated as follows:

$$x_c = \frac{\sum_{x=1}^{W} \sum_{y=1}^{H} x\, M(x,y)}{\sum_{x=1}^{W} \sum_{y=1}^{H} M(x,y)}, \qquad y_c = \frac{\sum_{x=1}^{W} \sum_{y=1}^{H} y\, M(x,y)}{\sum_{x=1}^{W} \sum_{y=1}^{H} M(x,y)} \tag{1}$$

The resulting centre coordinates are represented as:

$$C_{\mathrm{ROI}} = (x_c, y_c) \tag{2}$$

Preprocessing is also applied to full mammograms. Varoquaux and Cheplygina emphasise the effect of human-generated artefacts on model predictions [3]. Usually, these annotations are handwritten on the non-breast area. If included in the data, the model could learn these patterns and develop biases as a result when classifying new inputs. Although such artefacts may contain predictive information, they are highly correlated with radiologists’ methods, places or periods. Annotations are removed using OpenCV. Thresholding separates foreground structures from the background, while morphological operations reconnect breast tissue and remove small isolated patterns. Connected-component analysis then retains the most plausible breast region and removes the remaining artefacts. Finally, the image is cropped according to the target aspect ratio and resized to the model input dimensions, while limiting geometric distortion and tissue loss.

Pipeline structure

The pipeline structure is explained in Figure 1.

Architecture

The project will use only the CBIS-DDSM dataset. No other source will be explored. The limited dataset motivates the use of controlled baselines, regularisation and progressively increased model complexity.

The literature makes a case to use dual inputs to refine the patient diagnosis. Abdikenov et al. uses both CC and MLO views to classify the abnormality as benign or malignant. Unfortunately, it significantly reduces the number of cases for training. CBIS-DDSM dataset sometimes has only one view for a patient and, assuming they are both available, they would need to be paired to account

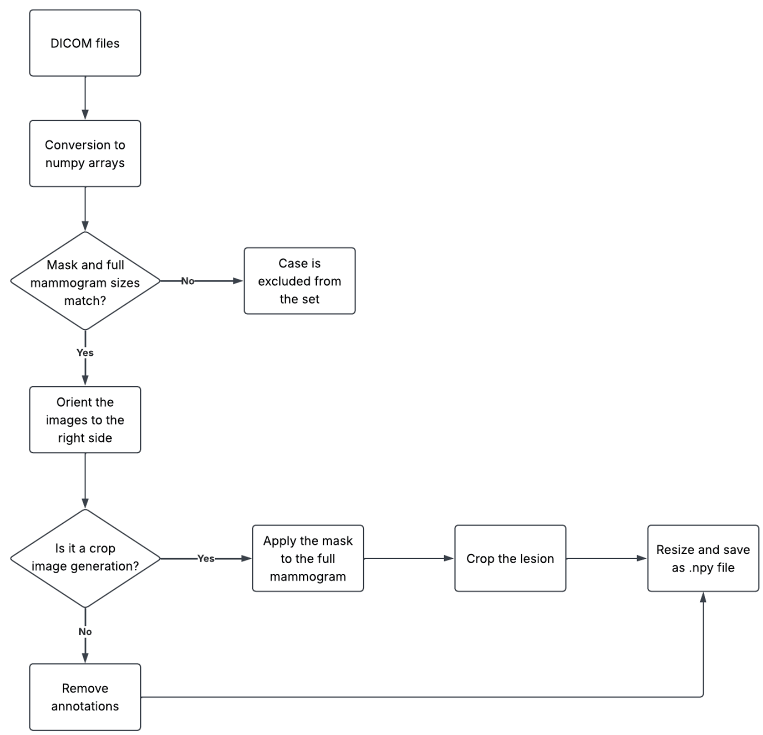

Figure 1: Pipeline

for one input. CNNs require large datasets to properly learn patterns. This project supports a two-level design in which global context and local lesion detail are treated as complementary sources of information.

This design is inspired by GMIC architecture, which combines a global module operating on high-resolution mammograms with a local module focused on selected informative regions [8]. However, the present project is narrower in scope, because it focuses on CADx classification of known abnormalities only. The proposed pipeline therefore trains two complementary branches: a global branch using the full mammogram representation, and a local branch using ROI-centred patches. Their features will then be merged before the final benign/malignant classification layer.

A first basic CNN model will be built to fairly compare preprocessing approaches. The chosen output layer activation function is sigmoid. Indeed, the problem is a binary classification matter - Does the patient have a malignant tumour? The model outputs a value between 0 and 1 representing the estimated probability of malignancy. Defining the classification threshold separating malignant from benign predictions is experimental and will be achieved based on the desired sensitivity-precision trade-off, based on the testing conducted on the validation set.

The model's architecture and hyperparameters will be input-dependent. the GMIC "first applies a low-capacity, yet memory-efficient, global module on the whole image" [8]. The module is light, but the full mammogram input image has a high resolution of 2944 x 1920 pixels. Such an input resolution may be difficult to replicate in this project. It might be downsampled, depending on whether the computation phase exceeds the machine's capabilities. The cropped images, on the other hand, have a lower resolution, which can easily be handled in the present scope.

The two branch features will be merged. The fused model receives the feature vectors produced by the global and local branches. The concatenated feature representation is passed to the final classification layer to produce the malignancy prediction. The resulting architecture follows a two-branch structure:

Due to the lack of available data in CBIS-DDSM, this baseline may reach a performance ceiling, even after optimization. Improving it entails the use of transfer learning techniques. According to Wang et al., these techniques enable "DL models to be trained on smaller datasets by leveraging information from related tasks or domains. This approach holds particular promise in medical imaging, where data scarcity poses a significant obstacle." [1]. Candidate architectures include DenseNet121, EfficientNetB0 and ResNet50.

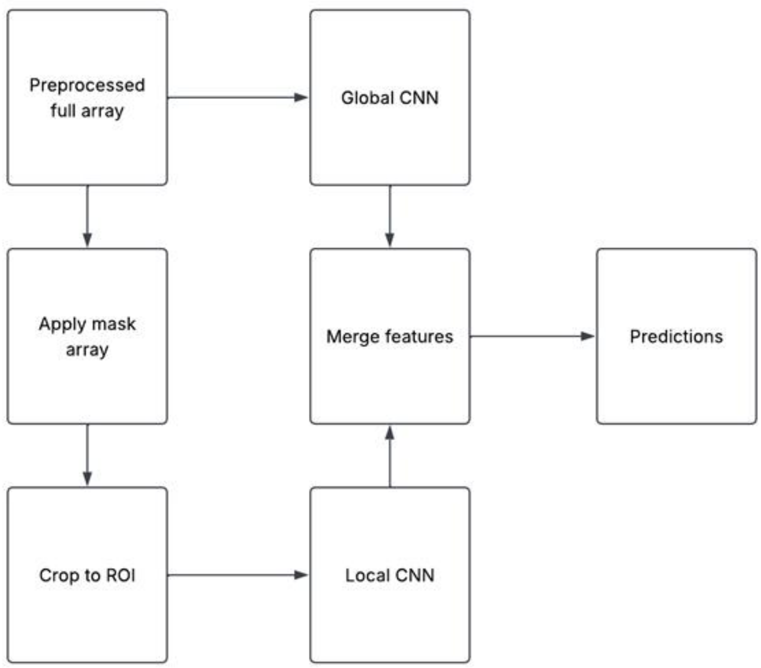

Figure 2: Architecture

Methods and Technologies

Dataset building

This step requires resolving the DICOM paths and verifying that no patient identifiers overlap across the splits. This is mostly data manipulation implemented using numpy, pandas, matplotlib and pydicom libraries. The latter helps identify the type of file from its SeriesDescription.

Numpy file generation

The design segregates the conversion of DICOM files from the model training, because it is too computational intensive. The numpy file generation converts the DICOMs into three-dimensional numpy arrays adapted to the chosen resolution and cropping criteria. It uses numpy, pandas, matplotlib, pydicom and tensorflow libraries. For the annotation removal, OpenCV is used.

Model training

Model training is implemented with tensorflow/keras, while scikit-learn is used to compute evaluation metrics. numpy, pandas and matplotlib support data handling and visualisation.

Grad-CAM

Grad-CAM visualisations are generated using numpy, pandas, matplotlib and tensorflow/keras.

Work plan

The project will be developed in four main phases: data preparation, representation selection, final modelling, and evaluation/reporting.

Task	**June**	**July**	**August**	**September**
Dataset ingestion and metadata matching	■			
ROI-mask alignment and visual checks	■			
Preprocessing comparison	■	■		
Select best image representation		■		
Train improved single-branch CNN		■	■	
Implement simplified global-local model		■	■	
Tune regularisation and augmentation			■	
Evaluation and interpretability outputs			■	■
Final report and figures			■	■

Table 2: Planned project timeline

Testing and evaluation

In the medical field, the risk of missing a malignant case is higher than flagging a false positive. Accuracy alone is not a sufficient metric. It needs to be complemented by recall and precision. These two metrics will be separately assessed. The trade-off will consist in increasing the precision while maintaining a high recall.

An extreme case would be that all classes are predicted as malignant. In such a case, the recall would be 1 and the precision would equal the proportion of malignant cases in the test set. Therefore, a relevant model needs to be capable of discriminating between benign from malignant cases with some validation performance. Maximizing ROC-AUC is thus prioritized, because it indicates whether the model can rank malignant cases above benign cases across classification thresholds. Over the validation phase, ROC-AUC is closely monitored. If validation ROC-AUC does not improve as the model runs on subsequent epochs, training is stopped after five consecutive epochs without improvement.

These metrics are useful, but defining whether the final model has improved requires a baseline definition. This project aims to determine whether a two-level architecture addresses the shortcomings of a model that relies exclusively on either the global or the local view. Before they are merged into a single module, the individual branches can be used as baselines. The dual-branch approach would then indicate whether the fusion of features improves the performance when compared to its single-branch counterparts.

Feature Prototype

Deep-learning is an experimental field. Conducting a successful experiment requires 1) a pipeline creation that handles data and train a model and 2) a virtuous cycle to gradually improve the model. The prototype creation aims at proving these matters can be addressed.

The feature prototype is organised into separate modules for dataset construction, preprocessing, model training, evaluation and explainability.

```
src/
|-- data/
|   |-- build_dataset.py
|   `-- train_val_test_split.py
|-- preprocessing/
|   |-- dataset_preprocessing.py
|   `-- dicom_handling.py
|-- training/
|   |-- dataset_preparation.py
|   |-- cnn_global_branch.py
|   |-- cnn_local_branch.py
```

```
|   `-- cnn_evaluation.py
|-- xAI_gradcam/
|   `-- grad_cam.py
|-- models/
|   |-- model_global_branch.keras
|   `-- model_local_branch.keras
|-- config.py
`-- functions.py
```

Inputs

Dataset creation

The CBIS-DDSM original package is made of DICOMs representing the mammograms and two CSV files - mass_case_description_train_set.csv and mass_case_description_test_set.csv - pointing to their location for every abnormality case.

Unfortunately, these paths are partially incorrect. The file names disclosed in the CSV do not match the actual one even though the given paths point to the correct folders. Building the dataset first requires to resolve the file names. For every case, for the fields "ROI mask file path", "image file path" and "cropped image file path" the files located in these paths are retrieved and their nature is assessed using pydicom.dcmread, which allows to retrieve the parameter SeriesDescription, stating the nature of the file. Once resolved, the actual paths are added to the dataset. Two fields are also added. "Label" converts the severity of the lesion into a binary. "BENIGN" and "BENIGN_WITHOUT_CALLBACK" are worth 0 and "MALIGNANT", 1. If the label returns N/A, the row is excluded.

Due to the risk of data leakage, the sets are fixed in three csv files - train_split, val_split, test_split. It also ensures the experiments are conducted under comparable conditions and can be reproduced. The train / validation split is made based on a traditional 80% / 20% ratio of the train set. There is a check conducted ex-post to confirm there is no patient overlap.

Preprocessing

The DICOMs are first saved as numpy file in a folder following a naming convention reflecting the preprocessing characteristics. Converting the files during the model training was considerably slowing the process, because each model is run three times with different seeds [42, 123, 999] to ensure the results are reliable. The DICOM files can be converted into arrays using pydicom library. The resulting files are two-dimensional arrays. Each breast image is oriented to the right side of the image to normalize the reading. It is achieved by converting the 2D array into a boolean one that returns True for every pixel exceeding a threshold value that defines whether the location may represent tissues. The side with tissue is pivoted to the right. The files need to be refined further to comply with the global or local branch type preprocessing.

Local inputs The mask is used to locate the tumour in the array. As a prerequisite, the mask and the image sizes are confirmed to be the same. If not, the case is excluded from the set, logged in a CSV file named "logs_history.csv" and a log is sent to the terminal for review. The total number of excluded cases is returned after the files were created.

Mask files that pass the check have the geometric center of the lesion calculated. A square 256 $\times$ 256-pixel crop is centred on the lesion centroid. The array needs to have a third dimension added to be used by the CNN. It is added using expand_dims, from the numpy library, and converted into tensor with tensorflow. The object is saved as .npy file under a newly created or reset folder.

Global inputs The mask is out of scope for global input arrays. However, it also is subject to preprocessing, because annotations need to be removed. A raw and fixed crop sometimes removed relevant breast tissue and was therefore considered unsuitable for the final preprocessing strategy. It was subsequently replaced by an OpenCV-based method combining thresholding, morphological operations and connected-component analysis. This refined approach aims to retain the main breast region while removing smaller isolated annotations and background artefacts.

Preprocessing review Although the size control caters to most risks, a visual control is needed ex-post to check possible, yet obvious, issues. Even though the project does not onboard trained radiologists, a spot check is still possible on plausible areas. For instance, does the mask overlap breast tissues? Is a lesion plausible in this localisation? The local check is expected to differ from the global one. The latter raises the question of whether relevant tissue context was removed. This project relies on the prerequisite that the CBIS-DDSM has been properly reviewed and that unusable files were ruled out.

Every time the DICOMs are converted into numpy (.npy), a series of 50 visuals saved as PNG is thus created. For each mammogram, it displays information converted as image.

- The raw full mammogram
- The raw mask, a black/white image. The white area is the lesion's
- The output numpy presented as image
- Full mammogram, no zoom, lesion highlighted, transparent mask applied. It helps defining whether the focused area is plausible
- Full mammogram, no zoom, lesion highlighted
- Zoomed mammogram, lesion highlighted

The preprocessed output corresponds to the third image. Such control helps identifying obvious mistakes in the preprocessing by ensuring that the end-of-chain mammogram is consistent with the desired characteristics. The generated NPY file matches the displayed image.

Numpy file dataset A CSV file is also created that resolves the paths of these newly created numpy files. There is one CSV per preprocessing type because they point to different folders depending on their preprocessing specifics.

CNN Model

The prototype tackles the training of the global and the local branch, individually. At this level, no fusion module has been created. It is a development that will be achieved later in the project. The prototype ensures that both branches manage to train their models, each one having its constraints.

The numpy files are used as inputs in a higher-capacity model. The global branch inputs should have a higher resolution than the local branch ones. The global branch is lighter. It uses higher-pixel images but starts with 16 filters, reducing the resolution from the first layer. This reduces computational cost when processing the larger global input while conserving the global context.

Global branch parameters

Parameter	Initial value
Input	Full mammogram
Input shape	$(768 \times 512 \times 1)$
Batch size	16
Maximum epochs	50
Convolutional blocks	3
Convolutional filters	16, 32, 64
Convolutional activation	ReLU
Feature aggregation	Global Average Pooling
Output activation	Sigmoid
Loss function	Binary cross-entropy
Model selection metric	Validation ROC-AUC
Early stopping patience	5 epochs

The local branch receives a zoomed 256x256-pixel image allowing the model to start with 32 filters. Indeed, the local branch can use greater representational capacity because its input is smaller and focused on the lesion.

Local branch parameters

Parameter	Initial value
Input	ROI-centred lesion crop
Input shape	$(256 \times 256 \times 1)$
Batch size	16
Maximum epochs	50
Convolutional blocks	3
Convolutional filters	32, 64, 128
Convolutional activation	ReLU
Feature aggregation	Global Average Pooling
Output activation	Sigmoid
Loss function	Binary cross-entropy
Model selection metric	Validation ROC-AUC
Early stopping patience	5 epochs

The training uses a validation set and stops five epochs after it reached its highest AUC. The epoch with the highest validation ROC-AUC is saved in a checkpoint keras file. The iteration's highest confidence is the reference model when the training is completed. The output layer's activation is a sigmoid function. The decision is binary. It associates a probability with the malignant outcome. Setting the threshold where this probability plays out is empirically managed.

Outputs

The output is two trained models based on the (numpy file, label) inputs. As a prototype, the CNN model is a simple 3-convolutional layers and one output layer (refer to the Design section).

Evaluating the output model entails 1) a review of the chosen metrics, and 2) an assessment as to whether the model learns useful patterns.

Metrics evaluation

During the validation phase, accuracy, precision and recall are computed at thresholds of 0.35, 0.40, 0.45 and 0.50. Because medical classification involves a trade-off between sensitivity and specificity, the decision threshold is selected on the validation set with particular attention to maintaining high sensitivity while preserving acceptable precision. ROC-AUC and the range of predicted probabilities are also reported.

Once selected, the threshold is fixed and applied unchanged to the test set. Any reduction in performance on the test set is interpreted as evidence of limited generalisation rather than used to redefine the decision threshold.

Model relevance

Ensuring that the DICOM inputs are processed as expected does not guarantee that the resulting NumPy arrays provide an appropriate representation for model training. In the global branch, full mammograms must be resized to match the target input dimensions. This operation may alter the original aspect ratio, potentially affecting the features learned by the CNN and, subsequently, their contribution to the fused representation. Visual spot checks are therefore used to identify excessive deformation, tissue loss, or other obvious preprocessing errors.

The project includes a Grad-CAM module to highlight the area leveraged by the model to learn. The idea is to initiate a feedback loop to adjust the preprocessing parameters. If it turns out that the model leverages patterns not related to the disease itself, some preprocessing features might have to be removed or added. For instance, initial Grad-CAM outputs suggested that the model was learning annotations located in the upper-left part of the mammogram. This observation motivated the replacement of the initial fixed-cropping approach with the OpenCV-based annotation-removal method described previously. Grad-CAM will continue to be used to assess whether the refined preprocessing directs the model towards clinically relevant image regions.

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